VERIFIED BIBLIOGRAPHIC & EMPIRICAL EXTRACT (EUROPE_TECH_UNIV_2023_2024)
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Title: Consolidated Financial Statements, Statistical Yearbooks, Tuition Schedules, and External Research Ecosystem Benchmarks: TUM, ETH Zurich, and Imperial College London (2023-2025)
Authors: TUM (TUM in Zahlen 2023/2024), ETH Zurich & ETH Board (Annual Reports 2023/2024), Imperial College London (Annual Report and Accounts 2023-24) (2024)
Source: Official University Financial Statements, Statistical Yearbooks & National Statistical Offices (Destatis, BFS, HESA)
DOI/URL: https://www.tum.de/ueber-die-tum/daten-und-fakten/tum-in-zahlen | https://ethz.ch/en/the-eth-zurich/portrait/information-material.html | https://www.imperial.ac.uk/about/governance/annual-report/
Sample: N = TUM: 52,580 (2023) / 52,931 (2024) students; ETH Zurich: 25,380 (2023) / 26,198 (2024) students; Imperial: 23,311 (2023-24) students

EXTRACTED EMPIRICAL METRICS & TABLE COORDINATES:

• TUM (2023/2024): Consolidated budget incl. Klinikum rechts der Isar = EUR 1,892.9M (2023: state core 734.5M, third-party 482.3M) rising to >EUR 2,300M (2024/25 with TUM Klinikum merger); Core academic budget excl. Klinikum = EUR 1,098.2M (2023: state core 601.3M, third-party 414.7M); Third-party funding (Drittmittel) >EUR 500M in 2024. Enrollment: 52,580 (2023) / 52,931 (2024); ~650-700 professors.
• TUM Garching & Munich External Research Ecosystem: MPI for Plasma Physics (IPP) EUR 146.6M; MPI for Extraterrestrial Physics (MPE) ~EUR 45M; MPI for Quantum Optics (MPQ) ~EUR 35M; MPI for Astrophysics (MPA) ~EUR 20M; MPI of Biochemistry (Martinsried) ~EUR 85M (+EUR 500M Bavarian campus investment); Helmholtz Munich EUR 351.0M (incl. EUR 80.5M third-party); ESO Garching HQ EUR 145M-160M; FRM II / MLZ Neutron Source EUR 72.0M; Leibniz Supercomputing Centre (LRZ) ~EUR 65M operating (+EUR 514M compute upgrade). Connected via joint professorships (Doppelberufungen).
• ETH Zurich & ETH Domain (2023/2024): ETH Zurich Total Operating Revenue = CHF 1,956M (2023) / CHF 2,000M (2024) [Federal contribution CHF 1,365M (2023) / CHF 1,422M (2024); Research contributions/mandates CHF 383M (2023) / CHF 403M (2024); Tuition CHF 37M / CHF 40M]. Total Operating Expenses = CHF 1,942M (2023) / CHF 2,024M (2024). Enrollment (2024): 26,198 total (11,304 BSc, 9,517 MSc, 4,351 Doctoral), 567 professors (headcount), 10,680 total FTE staff (6,622 scientific FTE). ETH Domain 2024 Federal Allocations: PSI CHF 304M, Empa CHF 109M, Eawag CHF 64M, WSL CHF 63M.
• Imperial College London (2022-23 / 2023-24): Consolidated Income = GBP 1,268.8M (2022-23) / GBP 1,329.4M (2023-24) [up to GBP 1,492.6M in 2024-25]; Operational Expenditure = GBP 1,240.3M (2022-23) / GBP 1,327.2M (2023-24 excl. USS pension credit; GBP 1,082.5M statutory). Research grants & contracts >GBP 363M (heavily concentrated in Faculty of Medicine / clinical NHS trusts & Francis Crick Institute founding partnership). Enrollment (2023-24): 23,311 students (12,191 UG, 9,196 FT PG, 1,924 PT PG); 8,501 FTE staff.
• International Master's Tuition, Duration & Living Costs (2024/25-2025/26): Imperial MSc (1.0 yr / 12 mos): Overseas tuition GBP 39,000-43,000/yr + London living GBP 22,722/12 mos (GBP 424-441/wk). ETH Zurich MSc (1.5-2.0 yrs): International tuition tripled Autumn 2025 from CHF 730/sem (CHF 1,460/yr) to CHF 2,190/sem (CHF 4,380/yr) + Zurich living CHF 24,000-28,800/yr. TUM MSc (2.0 yrs / 4 sem): Non-EU tuition introduced WS 2024/25 at EUR 4,000-6,000/sem (EUR 8,000-12,000/yr) + Munich living EUR 14,400-18,000/yr.
• Post-Graduation STEM Master's Salaries: Zurich (BFS EHA / ETH): CHF 85,000-110,000 starting (CHF 100k-120k CS/EE), rising to CHF 115,000-150,000+ at 5-10 yrs. Munich (IG Metall ERA 10-12 / StepStone): EUR 52,000-62,000 starting, rising to EUR 85,000-105,000 at 10 yrs. London (Imperial Graduate Outcomes): GBP 37,000-60,000 starting (higher in quant/finance), rising to GBP 65,000-95,000 at 10 yrs.

VERBATIM SOURCE QUOTE / TABLE RECORD:

TUM in Zahlen 2023: Etat TUM ohne Klinikum: 1.098,2 Mio. EUR (davon 601,3 Mio. EUR Staatszuschuss, 414,7 Mio. EUR Drittmitteleinnahmen); Etat TUM mit Klinikum: 1.892,9 Mio. EUR (davon 734,5 Mio. EUR Staatszuschuss, 482,3 Mio. EUR Drittmitteleinnahmen). ETH Zurich Annual Report 2024: Total operating revenue CHF 2,000 million; Total operating expenses CHF 2,024 million; 26,198 students (11,304 Bachelor, 9,517 Master, 4,351 Doctoral). Imperial College Annual Report 2023-24: Total income GBP 1,329.4 million; 23,311 students; 8,501 FTE staff.